# Power Electronics Laboratory
# Session 3: Isolated DC-DC Converters

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## Introduction

This experiment investigates the operation and theoretical characteristics of two isolated DC-DC converters: the flyback converter and the forward converter. In the first part, we analyze the flyback converter in both Continuous Conduction Mode (CCM) and Discontinuous Conduction Mode (DCM). By testing different duty cycles and load steps, we quantitatively verify the output voltage equations and observe the zero-crossing behavior of the magnetizing current. In the second part, we study the forward converter in CCM. We specifically focus on the transformer's reset mechanism, which uses a third winding and a diode to safely discharge the core energy. Finally, by applying an extreme duty cycle step, we mathematically and visually prove the operational limits of the reset circuit and demonstrate how insufficient OFF-time leads to magnetic saturation.

# Part 1:

# 1 Task 1: Continuous Conduction Mode

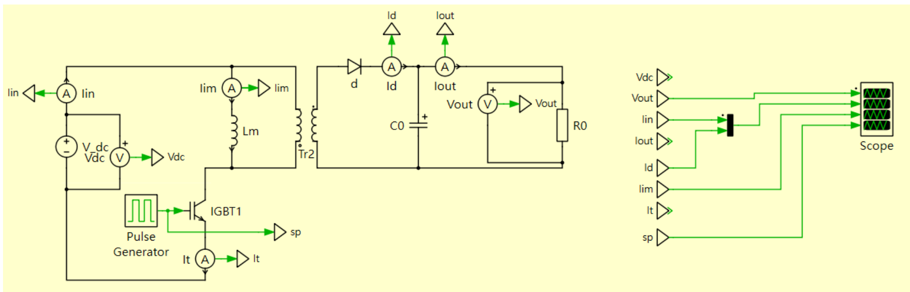


Figure 1: CCM circuit diagram

## 1.1 Duty Cycle Selection and Verification of CCM

To evaluate the flyback converter, three specific duty cycles ($D$) were selected to achieve output voltages below, equal to, and above the 100V input voltage: $D = 0.3$, $D = 0.5$, and $D = 0.7$, respectively. To verify CCM operation, the magnetizing current ($i_{lim}$) was examined across all three cases. For $D = 0.3$, $D = 0.5$, and $D = 0.7$, the minimum valley values of the magnetizing current are strictly greater than zero (13.8A, 46.5A, and 190.5A, respectively). Because the magnetizing current never returns to 0A before the next switching cycle begins, the converter is definitively operating in Continuous Conduction Mode.

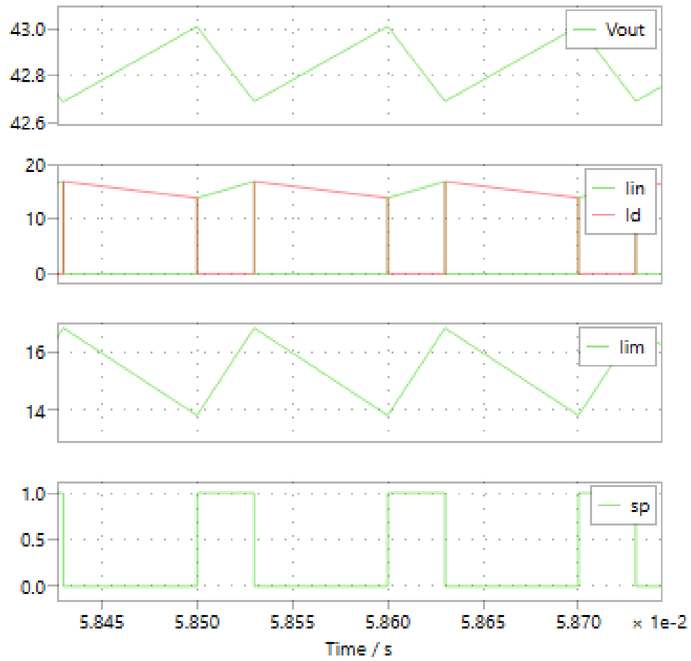


Figure 2: CCM (D=0.3)

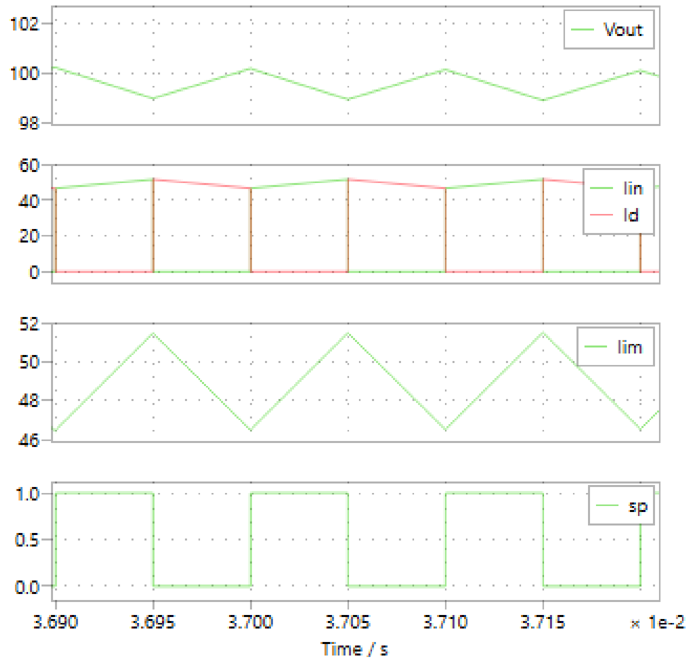


Figure 3: CCM (D=0.5)

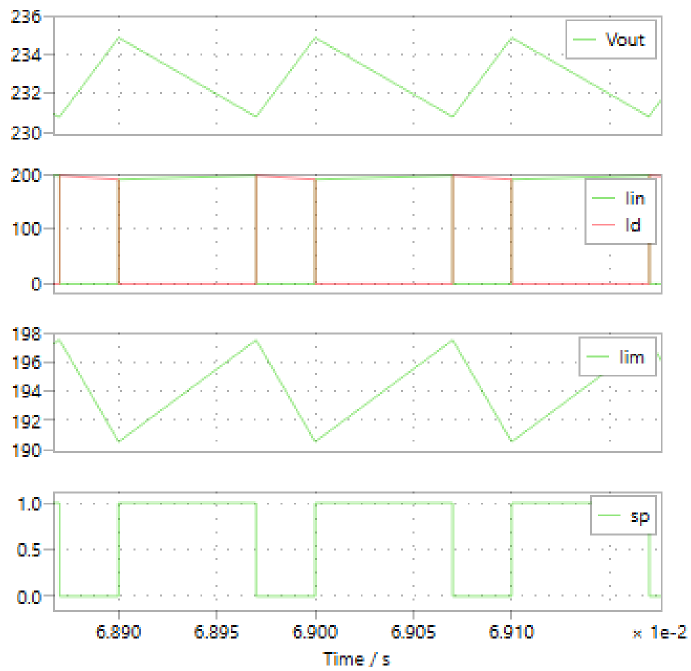


Figure 4: CCM (D=0.7)

### 1.2 Output Voltage ($V_{out}$) Analysis

The theoretical output voltage for a flyback converter in CCM is governed by the transfer function $V_{out} = V_{in}\frac{D}{1-D}$.

- Below $V_{in}$ ($D = 0.3$): The theoretical value is $100 \cdot \frac{0.3}{0.7} \approx 42.86$V. The measured average voltage from the plot is approximately 42.85V (fluctuating tightly between 42.7V and 43.0V).

- Equal to $V_{in}$ ($D = 0.5$): The theoretical value is $100 \cdot \frac{0.5}{0.5} = 100$V. The measured average is 99.6V.

- Above $V_{in}$ ($D = 0.7$): The theoretical value is $100 \cdot \frac{0.7}{0.3} \approx 233.33$V. The measured average is 232.8V.

- Comment: All measured output voltages match the theoretical calculations accurately. The minor deviations (e.g., 0.4V drop at $D = 0.5$) act as physical proofs of the slight energy losses caused by the 1m$\Omega$ transistor on-resistance and diode forward voltage.

### 1.3 Input Current ($i_{in}$) Analysis

In a flyback converter, the input source only supplies current when the primary switch is closed ($s_p = 1$).

- As shown in all three plots, $i_{in}$ perfectly mirrors the magnetizing current ($i_{lim}$) during the ON-interval and immediately drops to 0A during the OFF-interval.
- The peak input current quantitatively increases with the duty cycle, matching the peak magnetizing current: 16.8A at $D = 0.3$, 51.5A at $D = 0.5$, and scaling massively to 197.5A at $D = 0.7$. This proves that higher voltage gains in CCM demand exponentially larger peak currents from the input source.

### 1.4 Magnetizing Current Waveform and Ripple ($\Delta i_m$) Analysis

The theoretical magnetizing current ripple is calculated using the inductor voltage equation during the ON-state: $\Delta i_m = \frac{V_{in}}{L_m} D T_s$. With $V_{in} = 100$V, $L_m = 1$mH, and $T_s = 100\mu$s, the formula simplifies to $\Delta i_m = 10 \cdot D$.

- For $D = 0.3$, theoretical ripple = 3A. Measured ripple: 16.8A − 13.8A = **3**A.
- For $D = 0.5$, theoretical ripple = 5A. Measured ripple: 51.5A − 46.5A = **5**A.
- For $D = 0.7$, theoretical ripple = 7A. Measured ripple: 197.5A − 190.5A = **7**A.
- Comment: The measured peak-to-peak ripples perfectly align with the theoretical values across all tested duty cycles. This serves as definitive quantitative proof that the rate of change of the magnetizing current is purely dictated by the input voltage and the fixed magnetizing inductance.

# 2 Task 2: Discontinuous Conduction Mode (DCM)

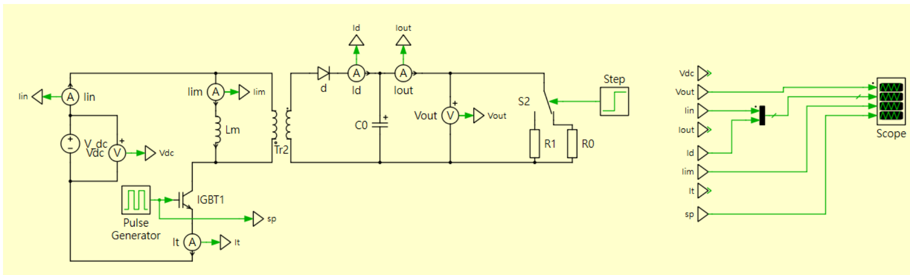


Figure 5: DCM circuit diagram

## 2.1 Setup and Transient Operation

To force the flyback converter into Discontinuous Conduction Mode (DCM), the duty cycle was kept constant at $D = 0.5$, and a load step was applied at $t = 0.50$s. The load resistance was increased from approximately $4\Omega$ (heavy load) to $200\Omega$ (light load) by opening a parallel switch. The transient response (Figure) shows the system transitioning from CCM to a new DCM steady state. To comply with the requirement of reducing the observed transient time, the subsequent analyses utilize highly zoomed-in plots (3 switching cycles, $\Delta t = 300\mu$s) to precisely evaluate the steady-state dynamics.

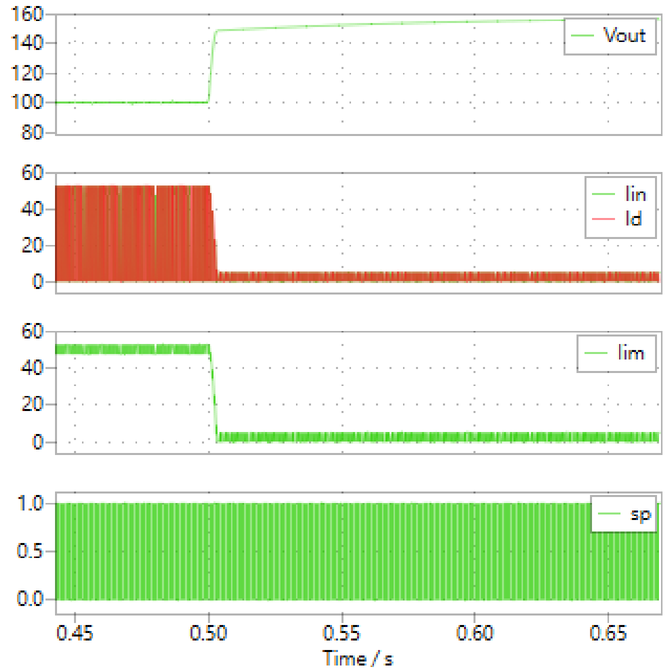


Figure 6: DCM (D=0.5)

## 2.2 Identifying the Zero-Crossing Instant

In the zoomed-in DCM (plot), the switching period is $T_s = 100\mu$s ($f_s = 10$kHz). With $D = 0.5$, the switch turns off at $t = 0.54795$s. The magnetizing current ($i_m$) peaks at 5A and then ramps down through the diode. As observed in the plot, $i_m$ reaches exactly 0A at approximately $t = 0.547983$s and remains at zero for the rest of the switching period (until $t = 0.54800$s).

- Quantitative Proof: The theoretical diode conduction time is $D_2T_s = \frac{L_m \cdot I_{m,peak}}{V_{out}} = \frac{1\text{mH}\cdot 5\text{A}}{152.06\text{V}} \approx 32.8\mu$s. This leaves a zero-current dead time of exactly $50\mu\text{s} - 32.8\mu\text{s} = 17.2\mu$s before the next cycle begins, which perfectly matches the visual data in the plot.

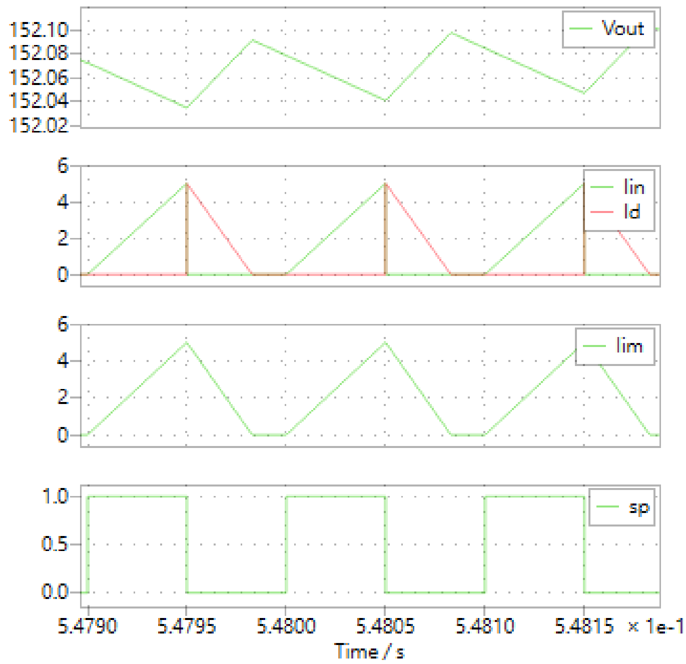


Figure 7: Enlarge figure of DCM D=0.5

## 2.3 Comparison: Output Voltage Variation

- CCM Operation: As shown in plot (CCM D = 0.5), the average output voltage ($V_{out}$) is regulated at 99.6V (fluctuating between 99V and 100.2V), which aligns with the theoretical CCM transfer function: $V_{in} \cdot \frac{D}{1-D} = 100\text{V} \cdot \frac{0.5}{0.5} = 100\text{V}$.

- DCM Operation: After the load step, the average $V_{out}$ drastically increases to 152.06V.

- Justification: In DCM, the voltage gain becomes load-dependent. The theoretical DCM output voltage is calculated as $V_{out} = V_{in} \cdot D\sqrt{\frac{R}{2L_m f_s}} = 100 \cdot 0.5\sqrt{\frac{200}{2 \cdot 10^{-3} \cdot 10^4}} \approx 158.1\text{V}$. The measured 152.06V is highly consistent with this theory, with the minor 6V discrepancy attributed to the forward voltage drop of the diode and the 1m$\Omega$ transistor on-resistance.

## 2.4 Comparison: Current Waveforms

- Magnetizing Current ($i_m$): In CCM, $i_m$ is continuous, operating with a massive average current of 49.0A and a peak-to-peak ripple of 5A (ranging from 46.5A to 51.5A). In DCM, the average $i_m$ drops significantly to accommodate the lighter load. It starts from 0A every cycle, reaching a maximum peak of exactly 5A ($I_{peak} = \frac{V_{in} \cdot D \cdot T_s}{L_m} = \frac{100 \cdot 0.5 \cdot 10^{-4}}{10^{-3}} = 5$A), completely eliminating the 46.5A DC bias seen in CCM.

- Input and Diode Currents ($i_{in}$ and $i_d$): In both modes, $i_{in}$ only flows during the ON-state ($s_p = 1$) and $i_d$ only flows during the OFF-state ($s_p = 0$). However, in DCM, $i_d$ strictly returns to 0A prematurely, leaving an idle period where neither the switch nor the diode conducts, confirming the complete discharge of the transformer's stored energy.

# Part2

# 3 Task 1: Duty Cycle Analysis

## 3.1 Simulation Setup and CCM Verification

Three duty cycles were selected to analyze the Forward converter: $D = 0.3$, $D = 0.5$, and $D = 0.7$. To verify that the converter operates in Continuous Conduction Mode (CCM), the output inductor current ($i_L$) was analyzed. Across all three cases, the minimum valley of $i_L$ remains strictly greater than zero (e.g., 12.9A at $D = 0.3$ and 32.8A at $D = 0.7$). Since the inductor current never reaches 0A, the system is successfully operating in CCM.

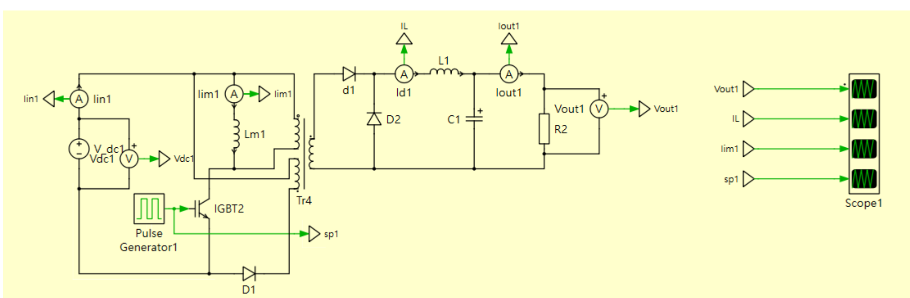


Figure 8: Forward converter circuit diagram

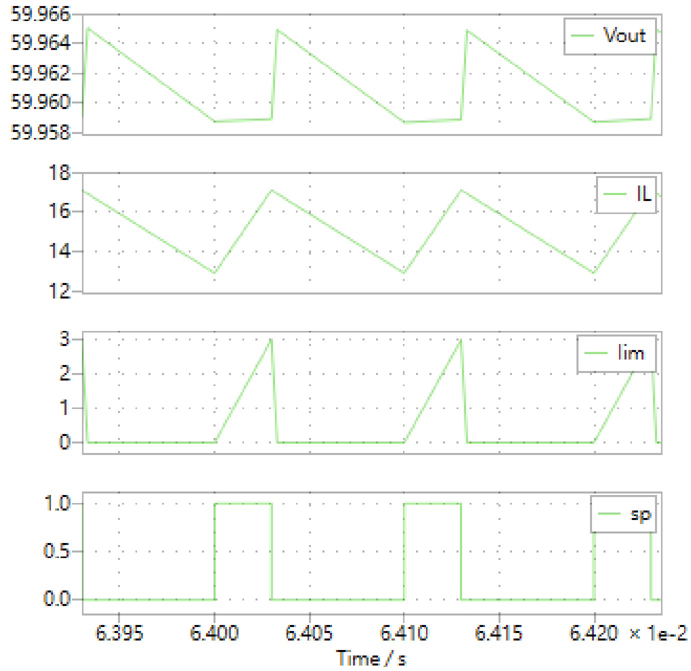
59.966
59.964
59.962
59.960
59.958
Vout
18
16
14
12
IL
3
2
1
0
lim
1.0
0.5
0.0
sp
6.395
6.400
6.405
6.410
6.415
6.420
× 1e-2
Time / s

Figure 9: D=0.3

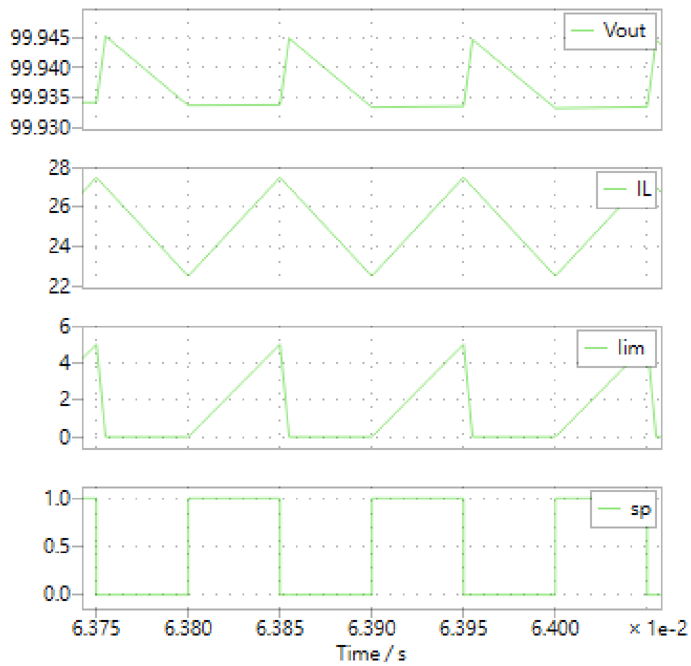


Figure 10: D=0.5

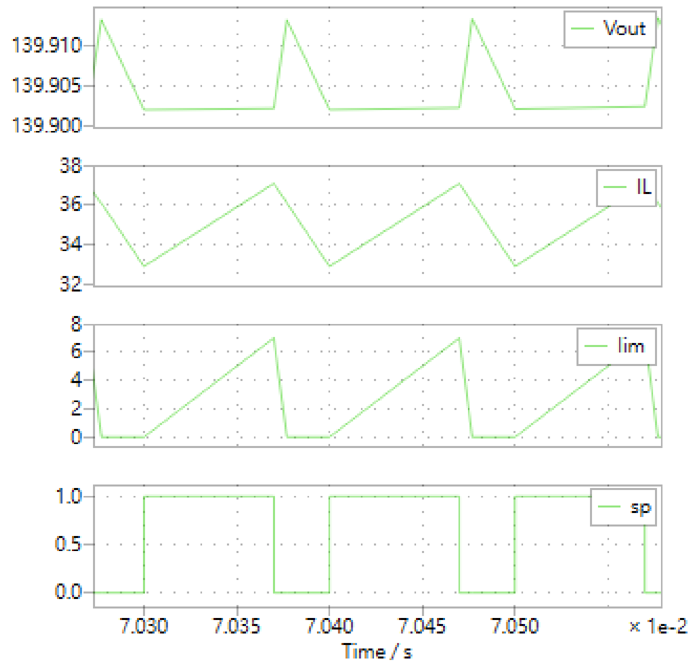


Figure 11: D=0.7

### 3.2 Output Voltage ($V_{out}$) Analysis

For a Forward converter in CCM, the theoretical output voltage is determined by the equation $V_{out} = V_{in} \cdot \frac{N_s}{N_p} \cdot D$. With $V_{in} = 100$V, $N_p = 1$, and $N_s = 2$, the expected voltage simplifies to $V_{out} = 200 \cdot D$.

- For $D = 0.3$: Theoretical $V_{out} = 60$V. Measured average $= 59.97$V.
- For $D = 0.5$: Theoretical $V_{out} = 100$V. Measured average $= 99.95$V.
- For $D = 0.7$: Theoretical $V_{out} = 140$V. Measured average $= 139.91$V.
- Comment: The measured output voltages perfectly match the theoretical predictions. The negligible drop of $< 0.1$V quantitatively proves the high efficiency of the circuit, accounting only for the minimal 1m$\Omega$ switch on-resistance and diode forward voltage.

### 3.3 Inductor Current ($i_L$) Analysis

The average inductor current theoretically equals the load current ($I_L = V_{out}/R$). With $R = 4\Omega$, the expected averages are 15A ($D = 0.3$), 25A ($D = 0.5$), and 35A ($D = 0.7$). The plotted center values align flawlessly with these figures.

- Ripple Proof: The theoretical peak-to-peak ripple is $\Delta I_L = \frac{V_{out}(1-D)}{Lf_s}$. For $D = 0.5$, $\Delta I_L = \frac{100\cdot 0.5}{10^{-3}\cdot 10^4} = \mathbf{5}$A. The plot identically shows $i_L$ fluctuating between 22.5A and 27.5A ($\Delta = 5$A), providing rigorous mathematical proof of the converter's output filter dynamics.

### 3.4 Magnetizing Current ($i_{lim}$) and Reset Circuit Analysis

The role of the third winding ($N_d = 0.1$) and diode $D_1$ is to demagnetize the core during the OFF-state.

- Current Peak: The theoretical peak magnetizing current is $I_{m,peak} = \frac{V_{in}\cdot D\cdot T_s}{L_m} = 10\cdot D$. The plots accurately display peaks of 3A, 5A, and 7A for $D = 0.3$, 0.5, and 0.7, respectively.

- Reset Verification: In all three plots, once the switch turns off, $i_{lim}$ ramps down steeply and reaches exactly 0A well before the next cycle begins, subsequently staying flat at 0A. Because $N_d/N_p = 0.1$, the reset time is extraordinarily fast ($t_{reset} = 0.1\cdot t_{on}$). This verifies that the reset mechanism operates correctly, safely preventing transformer core saturation.

## 4 Task 2: Reset Circuit Analysis

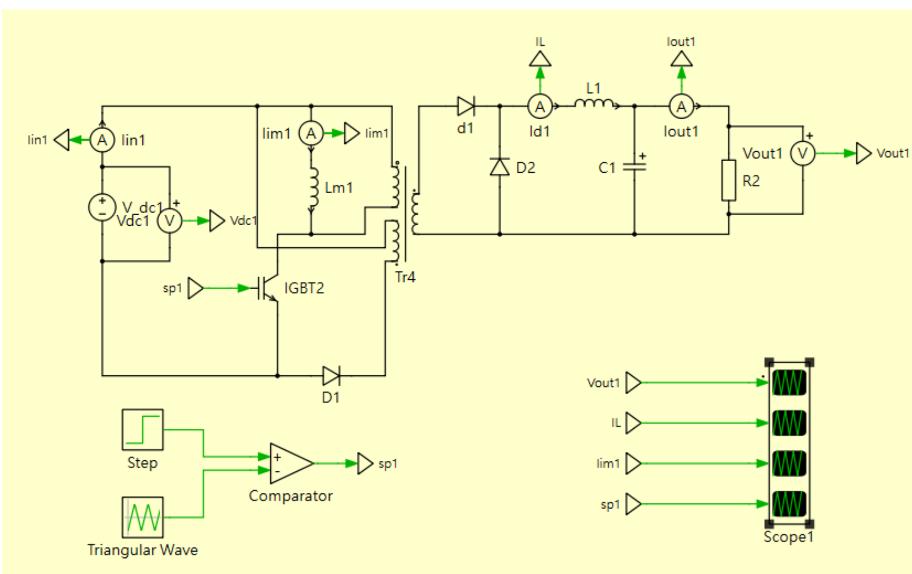


Figure 12: Forward converter circuit diagram

## 4.1 Duty Cycle Step Application

To analyze the limits of the reset circuit, a dynamic duty cycle step was applied. The system was initially set to $D = 0.5$ (where the reset mechanism operates perfectly) and then stepped to an extreme value of $D = 0.95$ at $t = 0.05$s. As observed in the resulting transient plot, after $t = 0.05$s, the magnetizing current ($i_{lim}$) fails to return to zero during the OFF-state. Instead, it increases sequentially in every switching cycle (staircasing effect), indicating that the core is rapidly approaching magnetic saturation.


Figure 13: Simulation output

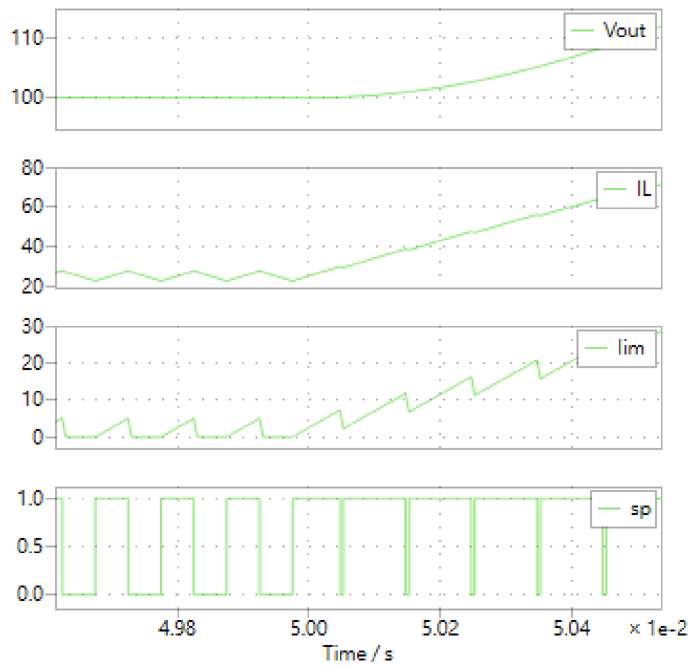


Figure 14: Enlarge around t = 0.05s

## 4.2 Analysis and Justification of Reset Failure

The failure of the reset mechanism at $D = 0.95$ can be mathematically justified by analyzing the current derivatives (slopes) and the available OFF-time:

- Charging Phase (ON-time): When the switch is closed, the primary voltage is clamped to the input voltage ($V_p = V_{in} = 100\text{V}$). The positive derivative of the magnetizing current is $\frac{di_m}{dt} = \frac{V_{in}}{L_m} = \frac{100\text{V}}{1\text{mH}} = 100\text{kA/s}$.

- Resetting Phase (OFF-time): When the switch opens, the reset winding ($N_d$) conducts through diode $D_1$. Because of the turns ratio between the primary and reset windings ($N_p/N_d = 1/0.1 = 10$), the primary voltage is forced to a high negative value: $V_p = -V_{in} \cdot (N_p/N_d) = -1000\text{V}$. The negative derivative for discharging is $\frac{di_m}{dt} = \frac{-1000\text{V}}{1\text{mH}} = -1000\text{kA/s}$.

- Time Required vs. Time Available: Since the discharging slope is exactly 10 times steeper than the charging slope, the time required to completely reset the core is $t_{reset} = 0.1 \cdot t_{on}$.

For the core to reset safely, the required reset time must be strictly less than or equal to the available OFF-time:

$$t_{reset} \leq t_{off}$$

$$0.1 \cdot (D \cdot T_s) \leq (1 - D) \cdot T_s$$

$$0.1D \leq 1 - D \implies 1.1D \leq 1 \implies D_{max} \approx 0.909$$

- Quantitative Conclusion: The theoretical maximum duty cycle for this specific transformer configuration ($N_d = 0.1$) is $D = 0.909$. By applying a step to $D = 0.95$, the ON-time becomes 95$\mu$s, requiring a reset time of 9.5$\mu$s. However, the available OFF-time is only 5$\mu$s (100$\mu$s $-$ 95$\mu$s). Since 5$\mu$s $<$ 9.5$\mu$s, the transformer simply does not have enough time to discharge its stored energy. This quantitatively proves why the magnetizing current accumulates in each cycle, causing the reset circuit to fail.

# Conclusion

Through this experiment, we successfully verified the theoretical principles of isolated DC-DC converters using quantitative simulation data. For the flyback converter, we proved that the output voltage accurately follows the $V_{in}\frac{D}{1-D}$ transfer function in CCM. We also demonstrated that forcing the system into DCM by increasing the load resistance causes the magnetizing current to fully discharge to 0A, which significantly increases the output voltage. For the forward converter, the measurements confirmed that the output voltage scales strictly with $V_{in}\frac{N_s}{N_p}D$. Most importantly, we validated the critical role of the reset winding. We mathematically proved that the transformer requires a specific minimum OFF-time to safely discharge its energy ($t_{reset} \leq t_{off}$). When we applied an extreme duty cycle of $D = 0.95$ (exceeding the theoretical maximum of 0.909), the available OFF-time was insufficient. This caused the magnetizing current to accumulate in every cycle, definitively proving the cause of reset circuit failure. Overall, this lab highlights the importance of proper duty cycle limits and load analysis in isolated power converter design.